

# Progress Report: Image-Constrained Generative Modeling of Retinal Vasculature

Ninghe Zhang

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## 1. Current Stage of the Project

This project studies a rule-based generative model for retinal vasculature under structural constraints extracted from retinal images. At the current stage, the implemented framework uses fundus images only. The present results should therefore be understood as a first-stage, fundus-constrained generative model rather than the full multimodal framework described in the proposal.

Compared with the previous update, the main progress this week is not in global constraint extraction, which had already been established, but in the density-aware branching rule. The earlier version mainly used local vessel density to adjust effective branching depth. The updated version now makes density act more directly on branch direction and branch survival, to improve where branches grow and where terminal nodes concentrate.

## 2. What Has Been Completed

The current pipeline includes three connected stages.

First, image-derived constraints are extracted from each fundus image. These include the retinal boundary, optic disc position and size, estimated macula position, vessel segmentation, vessel density map, and coarse major-vessel orientation. These components were already implemented in the earlier stage of the project and remain the basis of the current framework.

Second, three model variants are compared under the same pipeline: a baseline model with fixed manual parameters, an image-constrained model using extracted anatomical parameters, and a density-aware model that incorporates the vessel-density map into branching behavior. The baseline model has been preserved unchanged, so comparisons remain meaningful.

Third, the density-aware generator has now been revised in three ways. Local density adjusts effective branching depth, nudges branch direction toward higher-density regions, and affects branch survival probability after candidate branches pass geometric checks. This is the main modeling change in the latest update.

### 3. Main Results



Figure 1: Extracted structural constraints from representative fundus images, including optic disc location, macula estimate, and major vessel orientation.

Figure 1 shows the extracted structural constraints for three representative fundus images. The optic disc position, estimated macula location, and superior/inferior arcade orientations vary clearly across images. This confirms that the generator is no longer driven by a single default geometry. Instead, each image now provides its own anatomical starting conditions.

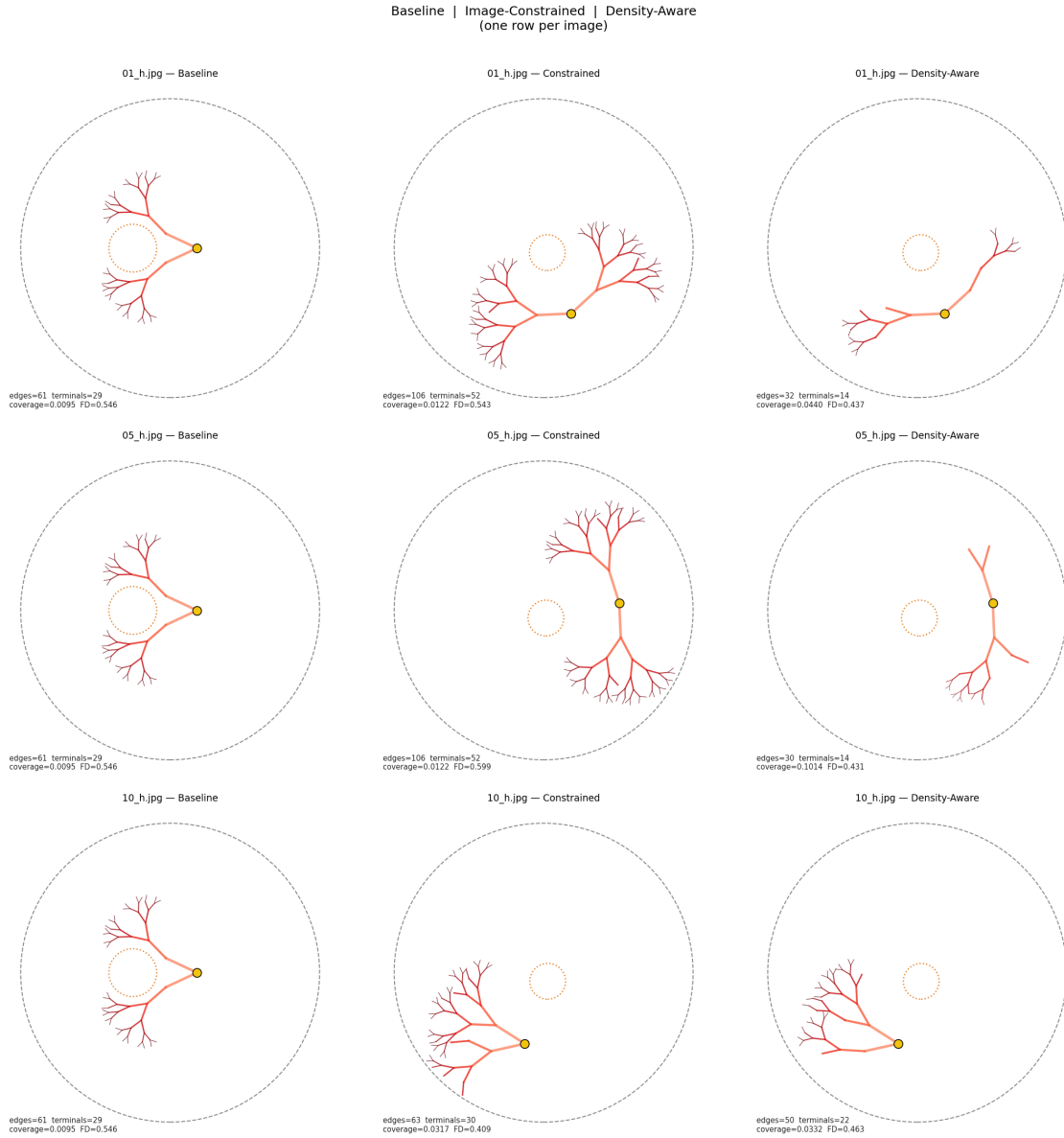


Figure 2: Comparison of baseline, image-constrained, and density-aware models across representative fundus images.

Figure 2 shows the qualitative comparison among the three model variants. The baseline model continues to produce nearly the same generic vascular tree across images. The image-constrained model, by contrast, shifts root placement and major branching direction according to the extracted fundus anatomy, producing visibly more image-specific structures.

The updated density-aware model behaves differently from last week’s version. It no longer simply adds branches while loosely following the constrained geometry. Instead, it produces a sparser but more selectively distributed tree. In the three representative examples, the density-aware model shows fewer edges and terminals than the image-constrained model, but the terminal nodes are placed in a way that appears more responsive to local density structure. This is especially visible in the first two images, where the updated density-aware trees are much less

diffuse than the constrained trees.

This indicates that the new density-guided direction and survival rules are active. The model is no longer using density only as an indirect depth modifier; density is now influencing where branches are allowed to persist.

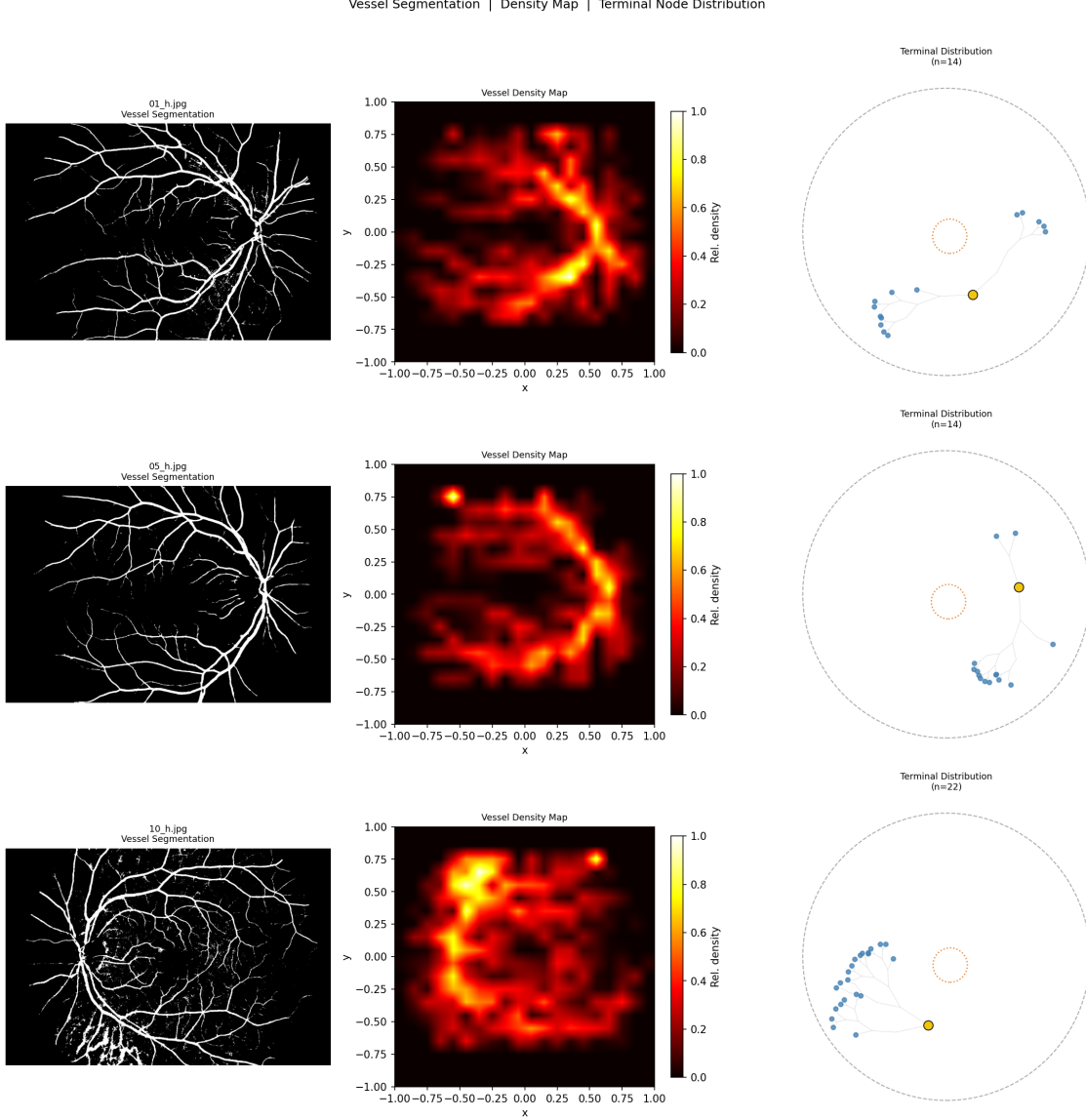


Figure 3: Vessel segmentation, density map, and terminal-node distribution for representative fundus images.

Figure 3 makes the density-aware update easier to interpret. The vessel segmentations and density maps show that real fundus images contain strong spatial asymmetry, with vessel structure concentrated in certain regions rather than spread uniformly across the retinal domain.

The terminal-node distributions from the updated density-aware model now reflect this asymmetry more clearly than before. Compared with the earlier stage, terminal nodes are less uniformly scattered and more visibly concentrated in selected regions of higher vessel density. In this

sense, the latest update addresses a weakness identified in the previous progress report, where density affected branching depth but did not directly guide where branches should go.

At the same time, the figure also shows that the current density-aware model may now be somewhat aggressive in pruning. In some images, the generated tree becomes quite sparse, with a relatively small number of terminals. This suggests that the revised branch-survival rule is effective, but may still require calibration to avoid over-pruning.

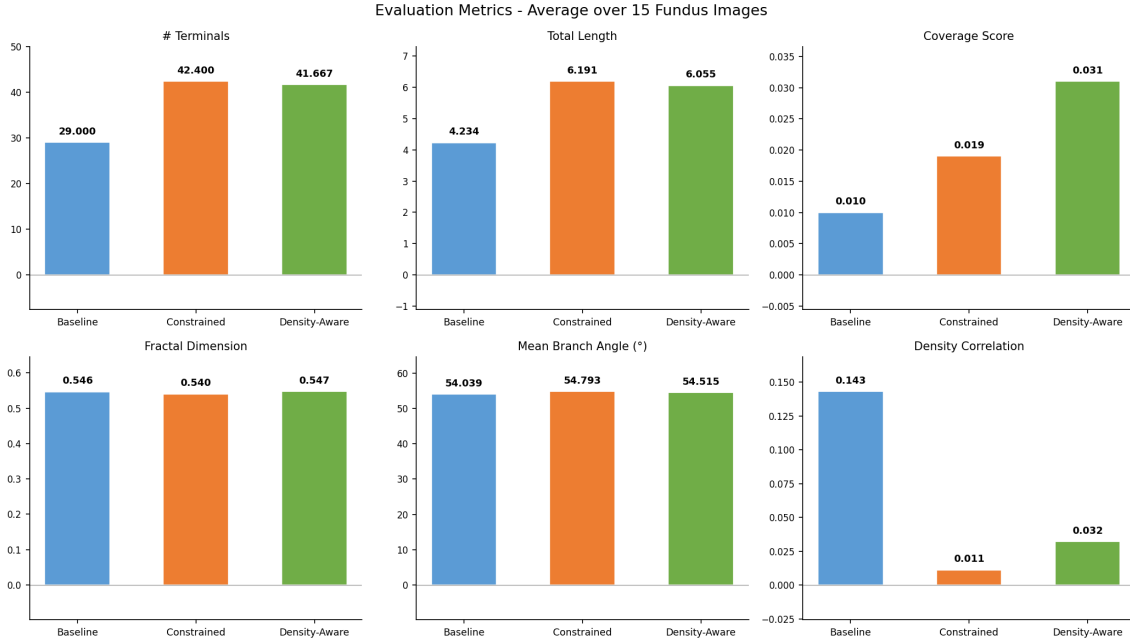


Figure 4: Quantitative evaluation metrics averaged over 15 fundus images.

Figure 4 summarizes the model comparison averaged over all 15 fundus images. This is a substantial improvement over the earlier stage, where the summary was based only on three representative images.

The results show three main patterns. First, both constrained models remain substantially richer than the baseline in structural complexity. The average number of terminals increases from 29.0 in the baseline to 42.4 in the image-constrained model and 41.7 in the density-aware model. Total vessel length also rises from 4.234 in the baseline to 6.191 in the constrained model and 6.055 in the density-aware model. This confirms that image-derived constraints continue to generate larger and more anatomically extended trees than a fixed manual baseline.

Second, the strongest improvement in the new update is in coverage. The coverage score increases from 0.010 in the baseline to 0.019 in the image-constrained model, and then to 0.031 in the density-aware model. This is the clearest sign that the revised density-aware mechanism is doing useful work: the updated model achieves substantially better coverage than both the baseline and the image-constrained model while using slightly fewer terminals and slightly shorter total length than the constrained model.

Third, density correlation improves relative to the previous density-aware result, but it remains

weak in absolute terms. The updated 15-image result is 0.032, which suggests that the new direction and survival rules are making terminal placement more responsive to the density map. However, the value is still low, and the baseline score remains unexpectedly higher. This means that density matching is still not fully resolved and should remain a priority for further investigation.

The remaining metrics are relatively stable. Fractal dimension stays close across all three models, and mean branch angle remains around 54–55 degrees, which suggests that the main changes are happening in spatial distribution and pruning behavior rather than in large angular distortions.

#### **4. Comparison with the Previous Stage**

Relative to the previous progress report, the current stage shows a clearer and more meaningful role for the density-aware model. Previously, the density-aware extension had only a modest effect and mainly altered effective branching depth. The main criticism at that stage was that density did not directly guide branch placement or terminal concentration.

The latest results show that this issue has been partly addressed. The density-aware model is now more selective, more spatially structured, and quantitatively stronger on coverage. The increase in average coverage from the earlier summary to 0.031 in the current 15-image evaluation is the strongest evidence of improvement. At the same time, the updated model also introduces a new tradeoff: better coverage is accompanied by more aggressive pruning and only modest density-correlation gain.

In other words, the project has moved from a density-aware model that was too weak to one that is now directionally effective but still not fully calibrated.

#### **5. Interpretation**

First, the project has now established a working fundus-based constrained generative framework. The model is no longer generic: it is anatomically anchored to each input image through optic disc location, macula exclusion, and major vessel orientation.

Second, the latest density-aware update is a real modeling improvement. It produces the strongest average coverage result so far and makes terminal placement more responsive to image-derived spatial structure.

Third, the main remaining weakness is not global anatomy but local density matching. The model now responds to density, but the density-correlation metric shows that this response is still limited and may be partly offset by over-pruning.

## 6. Next Steps

The first priority is to tune the density-aware survival and direction rules so that the model can preserve the coverage improvement without becoming too sparse. The current version appears to have moved in the right direction, but branch pruning may be too strong in some images.

The second priority is to understand the density-correlation metric more carefully. Since the updated model improves terminal placement qualitatively and improves coverage quantitatively, but still shows only weak density correlation, the metric itself and its relationship to the generated graph should be reviewed.

The third priority is to expand evaluation beyond summary graph statistics. A stronger comparison between generated terminal distributions and image-derived vessel-density structure would make the improvement more interpretable.

The fourth priority remains the same as in the original proposal: once OCTA data become available, the model should be extended from the current fundus-only framework to the intended multimodal setup. That extension is still necessary for incorporating deeper microvascular structure directly rather than approximating it from fundus vessel segmentation alone.

## 7. Summary

In summary, the current stage completes a stronger second version of the fundus-based image-constrained model. The main new contribution is the revised density-aware branching rule, which now uses density to influence branch depth, direction, and survival. Relative to the previous stage, the updated model shows a clear gain in coverage and a more plausible concentration of terminal nodes in higher-density regions. The main remaining task is to improve local density matching without excessive pruning, and then extend the framework to OCTA-derived constraints when those data become available.